1. Objective

Monitor interior cabin climate and vehicle ride dynamics using STM32-NUCLEO-G070RB, and print the data over the OLED display for the monitoring purpose.

2. Hardware

- MCU: STM32-NUCLEO-G070RB dev board
- Climate sensor: DHT22 (temperature, humidity)
- IMU: MPU6050 (3-axis accelerometer and gyroscope)
- OLED 1.3” (SH1106 driver) Display
- Power: 5V via micro-USB (from 12V automotive adapter)

Pin Mapping

Signal	STM32-NUCLEO-G070RB Pin
DHT22 VCC	3.3V
DHT22 GND	GND
DHT22 OUT	PB4
MPU6050 VCC	3.3V
MPU6050 GND	GND
MPU6050 SCL	PB8 (shared with OLED)
MPU6050 SDA	PB9 (shared with OLED)
OLED VDD	3.3V
OLED GND	GND
OLED SCL	PB8 (shared with MPU6050)
OLED SDA	PB9 (shared with MPU6050)
Inbuilt User LED (LD4)	PA5

3. Functional Requirements

- Read temperature (°C) and humidity (%) from the DHT22 every 2 seconds.
- Read X, Y, Z acceleration from the MPU6050 (±4 g range, 100 Hz) and convert to G-force. At power-up, keep the board flat and still for 1 s so the readings are zeroed (X = 0 g, Y = 0 g, Z = 1 g).
- Map axes: X = lateral (cornering), Y = longitudinal (accel/braking), Z = vertical (bumps).
- Print the sensor data on the OLED with the heading "Vehicle Parameters Display".
- Keep all alert thresholds configurable as macros in `config.h` (values below).
- "Suddenly" means the axis value changes by at least the threshold within 100 ms. The original 15 g was replaced because 15 g is a crash-level force and would never trigger.
- Display "Unseen SpeedBreaker" if Z changes suddenly by 0.30 g or more.
- Display "Wide Turn" if X changes suddenly by 0.25 g or more.
- Display "Speeding" if Y changes suddenly by +0.25 g or more (detected from acceleration, so it indicates harsh acceleration, not actual speed).
- Display "Sudden Braking" if Y changes suddenly by −0.25 g or more.
- If any of the above 4 events occurs, show the event message immediately on the OLED while keeping the Temperature and Humidity values visible, and blink the built-in LED (LD4) every second (500 ms on, 500 ms off). The message and blinking stay for 2 s after the last event; if events overlap, the priority is Sudden Braking, Unseen SpeedBreaker, Wide Turn, Speeding.
- If temperature or humidity goes above its limit, show "!" next to the value.

Parameter	Demo threshold	How to trigger in the demo
Temperature high	32 °C	Hold the sensor between warm fingers
Humidity high	75 %	Breathe gently on the DHT22
Z(Unseen SpeedBreaker)	0.30 g	Quick tap or lift-and-drop
X (Wide Turn)	0.25 g	Quick flick left or right
+Y (Speeding)	+0.25 g	Quick push forward
−Y (Sudden Braking)	−0.25 g	Quick pull backward
Sudden window	100 ms	
Event display time	2 s	

For in-vehicle use, raise the G thresholds to about 0.4 to 0.5 g and the temperature and humidity limits to 45 °C and 80 %.

4. OLED Display format

Vehicle Parameters
Display

Temp: “value from DHT22” °C
Humidity: “: “value from DHT22” %
X:<x>
Y:<y>
Z:<z> g

5. Expected Use

Installed in a vehicle, streaming cabin comfort and ride data to a dashboard for driver behavior review or ride comfort tracking, with immediate alerts on extreme cabin or dynamics events.